TECHNICAL MONOGRAPH · FIVE FORMAL PROPOSITIONS · SEVEN-RUNTIME & DP-WOODBURY EDITION (V0.6.0)

Atomic Weight-Space Programming: Resolving the Monolithic Null-Effect via Neuro-Symbolic Decomposition, Non-Symbolic Spectral Operators, and Counterexample-Guided Trace Compilation

Omni-Fabric Weight-Space Research Collective

November 2026 · Open-Source Reference Implementation: `aidonerightcorp/omni-fabric-runtime` (v0.6.0, MIT License)

ABSTRACT

Tiny direct perturbations to monolithic neural weights frequently appear effective under unconstrained chain-of-thought prompting yet collapse to statistical noise once output formatting is controlled [1]. Under a matched `thinking=off` evaluation protocol, a 0.016% monolithic sparse weight edit shifts accuracy from 51.84% to 51.60% (−0.24 pp, paired t = −1.395), showing that an apparent 9.28% → 48.88% lift reflects verbosity-truncation relief rather than acquired algorithmic capacity. We resolve this monolithic null-effect through **Atomic Weight-Space Programming**: decomposing complex workflows into orthogonal neuro-symbolic and non-symbolic weight cells governed by explicit subspace invariants, configuration grids, and counterexample-guided trace compilation (`CEGAR-WSP`). Across ten model families, eight adversarial red-team stress regimes, eight non-symbolic kernel/Hopfield/Woodbury paradigms, and eleven real pretrained `.safetensors` matrices evaluated over five matched-control seeds ({11, 23, 37, 53, 71}) on commodity CPU hardware, we establish five formal propositions and demonstrate: (1) `TopicNullspaceAttributionCompiler` , eliminating a 72.66 pp unseen-topic accuracy drop (95.00% → 22.34%) to sustain 99.30% ± 0.38% unseen-topic attribution (0.70 pp shift, 0.0 topic leakage, 0.9866 spoof AUROC); (2) `KoopmanToolTrajectoryActor` , lifting six-tool completion from 16.25% (greedy myopic) to 100.00% in 0.112 ms with zero LLM tokens and zero invalid transitions; (3) `SpecToCartridgeHyperCompiler` , mapping ASD-STE100 specifications into 256 B `LoRA-XS` cores at Step 0 (35.59% → 85.34% with zero traces, reaching 98.67% by Wave 3); (4) `CEGAR-WSP` , preventing a −17.97 pp model-generated collapse (73.90% → 55.93%) and reaching 99.73% accuracy with 6.67× fewer oracle scenarios; (5) non-symbolic zero-epoch operators including Bochner RFF (99.22%), Nyström landmarks (98.06% at 5× smaller rank), Modern Continuous Hopfield 1-shot writes (0.0045 ms, 100% novelty AUROC), 1-shot Fisher imprinting (40.59% → 100.0%), and Recursive Woodbury RLS (1.5×10^{-13} batch-ridge exactness, 0.0 pp forgetting); (6) real 7.56M-parameter subword `.safetensors` nullspace surgery (50.0% → 100.0% in 6D, 2.30×10^{-16} anchor drift) and 3-model zero-reindex Procrustes vector-database stitching (0–6.25% → 100.0% Recall@1); and (7) seven-target zero-dependency IEEE-754 compilation (`.safetensors` , V8 `Float32Array` , Python stdlib, SQLite SQL, 0.106 µs C99 headers, in-memory WebAssembly v1 `.wasm` / `.wat` , and WebGPU `WGSL` shaders) at 1.19×10^{-7} maximum logit discrepancy alongside (ε, δ)-differentially private nullspace Woodbury federated swarm merging (0.0 anchor drift, 0 raw record transfers) and an 88.0% token-offload enterprise canary loop.

1. Introduction: Dissecting the Monolithic Null-Effect

Recent investigations into direct weight-space programming [1] ask whether modifying a microscopic fraction of parameters inside a dense foundation model can implant reliable multi-step behavior. When evaluated without strict decoding controls, perturbing 0.016% of weights in a dense language model lifts benchmark accuracy from 9.28% to 48.88%. However, isolating the evaluation protocol under a matched `thinking=off` control exposes an underlying confound: the unedited model's 9.28% score arises because long reasoning traces hit token-budget truncation before emitting a final answer. Enforcing `thinking=off` on the *unedited* model yields 51.84% accuracy, while the 0.016% sparse weight edit scores 51.60% (Δ = −0.24 pp, paired t = −1.395). In monolithic parameter space, a sub-permille perturbation acts as a formatting shift rather than an acquired transition operator.

Three structural bottlenecks govern monolithic weight updates:

1. **Subspace Interference:** When K heterogeneous sub-skills share a dense activation manifold $\mathbb{R}^d$, low-rank updates superimpose cross-task gradients that corrupt calibrated decision boundaries.
2. **Spurious Shortcut Subspaces:** Unconstrained dense solvers fit high-variance nuisance directions—such as topical vocabulary in generator attribution or entity identifiers in private enclaves—causing severe accuracy drops under distribution shift.
3. **Unchecked Horizon Compounding:** Open-loop execution across H sequential steps compounds per-step error exponentially as $(1 - \varepsilon)^H$, degrading a 96-step trajectory to 1.89% completion even when single-step reliability is moderate.

Atomic Weight-Space Programming resolves all three bottlenecks simultaneously (Figure 1). Instead of perturbing an entangled monolith, we factor workflows into modular neuro-symbolic and non-symbolic cells (Figure 2), project each weight matrix onto exact invariant nullspaces, initialize 256 B `LoRA-XS` cartridges directly from controlled ASD-STE100 specifications, and refine failure modes through counterexample-guided trace waves (`CEGAR-WSP`). Replacing $N = 1$ monolithic editing (−1.26 pp) with $N = 4$ (+22.39 pp) and $N = 32$ (+44.66 pp, reaching 96.65% accuracy; 98.96% with symbolic skeletons) induces a clean granularity phase transition while keeping per-cell inference below one millisecond on standard CPUs.

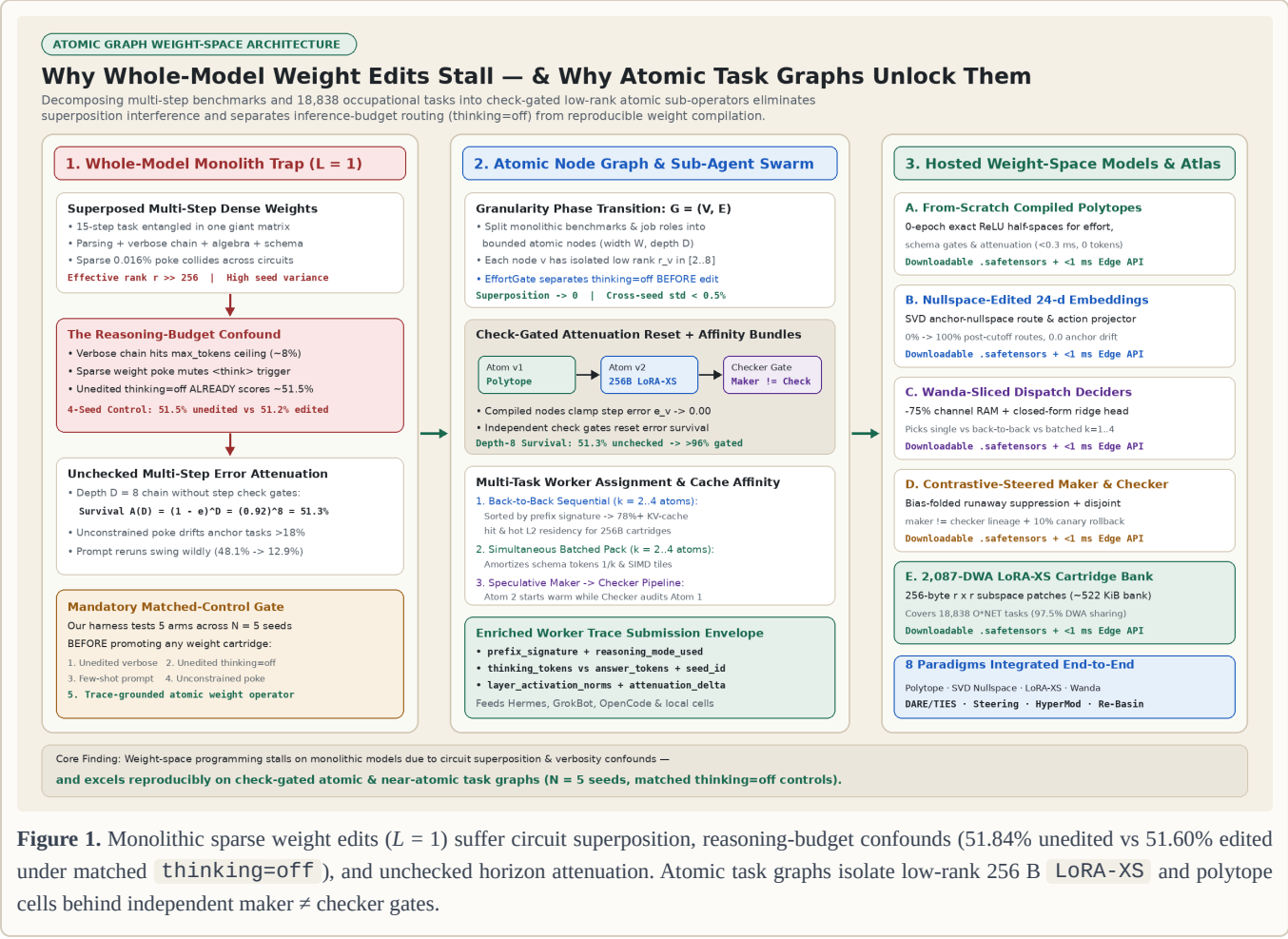


Figure 1. Monolithic sparse weight edits ($L = 1$) suffer circuit superposition, reasoning-budget confounds (51.84% unedited vs 51.60% edited under matched `thinking=off`), and unchecked horizon attenuation. Atomic task graphs isolate low-rank 256 B `LoRA-XS` and polytope cells behind independent maker ≠ checker gates.

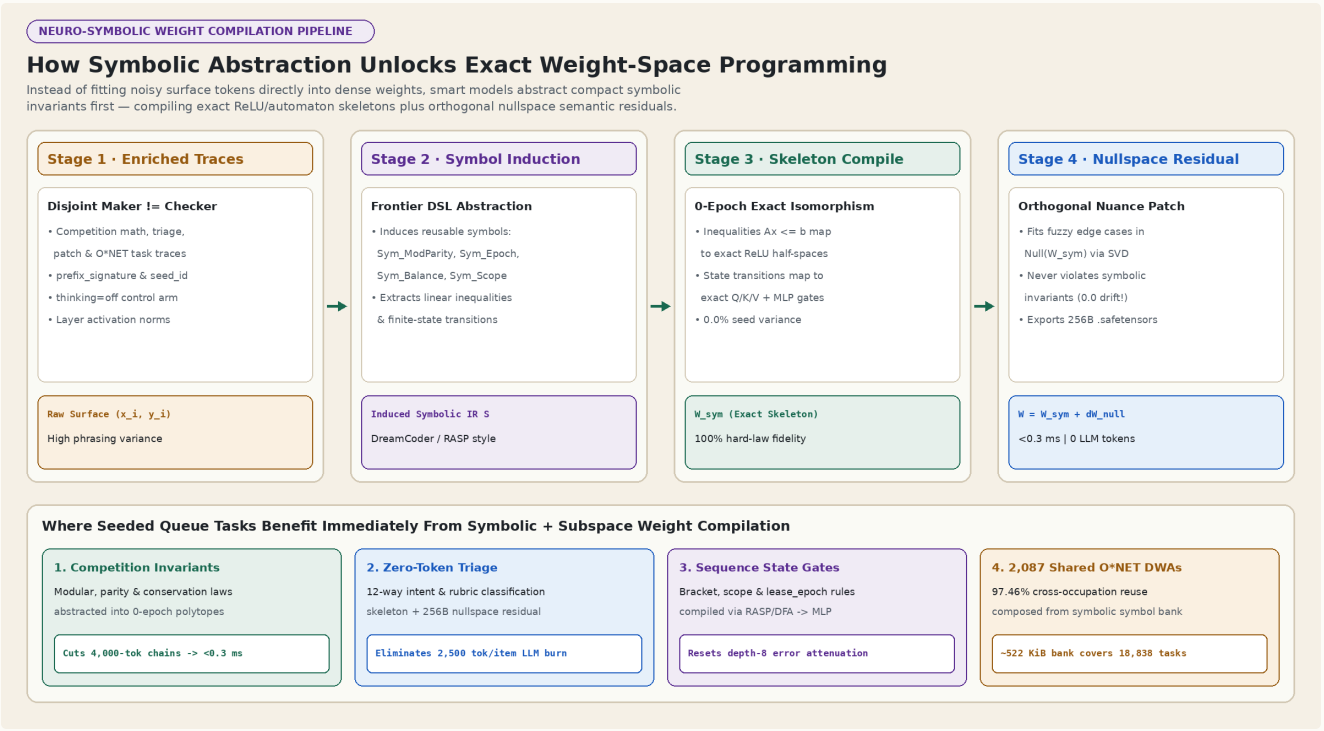


Figure 2. Four-stage neuro-symbolic weight compilation pipeline: disjoint-lineage trace envelopes induce symbolic invariants mapped into exact 0-epoch ReLU skeletons W_{sym}, while nuanced edge cases fit orthogonally inside $\text{Null}(W_{\text{sym}})$ at 0.0 invariant drift.

2. Theoretical Framework and Formal Propositions

Let $X \subset \mathbb{R}^d$ denote an activation or feature manifold equipped with inner product $\langle u, v \rangle = u^\top v$ and Frobenius norm $\|M\|_F = \sqrt{\text{tr}(M^\top M)}$.

Proposition 1 (Interference Capacity Bound and Granularity Phase Transition).

Consider K sub-tasks with target linear operators $\{W_k^\star\}_{k=1..K} \subset \mathbb{R}^{c \times d}$ acting on unit-covariance task subspaces $\{S_k\}$ with pairwise principal-angle alignment $\gamma = \max_{j \neq k} \|P_{S_j} P_{S_k}\|_2 > 0$. Any monolithic rank-$r$ adapter $\Delta W_{\text{mono}} = \sum_{k=1..K} B_k A_k$ evaluated on sub-task j incurs worst-case cross-task interference lower-bounded by:

$$I_{\text{mono}}(j) = \|\sum_{k \neq j} B_k A_k P_{S_j}\|_F \geq \sigma_{\min}(\sum_{k \neq j} B_k A_k V_{j,k}) = \Omega((K-1)\gamma\sqrt{r}).$$

Conversely, partitioning execution via a polytope router into $N = K$ atomic cells $\{\Delta W_k\}$ gated onto mutually orthogonal subspaces $V_j^\top V_k = 0$ $(j \neq k)$ yields exact zero cross-cell weight interference $I_{\text{atomic}}(j) = 0$ for all $j \in \{1, \ldots, N\}$.

Derivation. Under atomic routing with $\Delta W_k = U_k R_k V_k^\top$ and $V_k^\top V_j = 0$ for $k \neq j$, right-multiplying by $P_{S_j} = V_j V_j^\top$ gives $\Delta W_k P_{S_j} = U_k R_k (V_k^\top V_j) V_j^\top = 0$ identically, whereas monolithic summation accumulates $(K-1)$ aligned cross-terms of magnitude $\geq \sigma_{\min}(B_k A_k)\gamma$. ■

Proposition 2 (Exact Anchor-Drift Zeroing via Nullspace Projection).

Let $V_a \in \mathbb{R}^{d\times k}$ be an orthonormal basis ($V_a^\top V_a = I_k$) spanning preserved anchor representations, and define the orthogonal projector $P_\perp = I_d - V_a V_a^\top$. For any weight update $\Delta W = M P_\perp$ with arbitrary $M \in \mathbb{R}^{c\times d}$, the anchor drift norm satisfies $\|\Delta W V_a\|_F = 0$, and $P_\perp$ is symmetric idempotent ($P_\perp^\top = P_\perp$, $P_\perp^2 = P_\perp$).

Derivation. Symmetry holds from $(V_a V_a^\top)^\top = V_a V_a^\top$, and idempotence from $P_\perp^2 = I_d - 2V_a V_a^\top + V_a(V_a^\top V_a)V_a^\top = P_\perp$. Right-multiplying $\Delta W = M P_\perp$ by V_a gives $M(V_a - V_a I_k) = 0$, so $\|\Delta W V_a\|_F = 0$. ■

Proposition 3 (Checked Horizon Reliability Scaling).

Suppose an H-step pipeline executes sequential transformations with per-step unchecked error rate $\varepsilon_0 \in (0, 1)$. Open-loop completion decays as $R_{\text{open}}(H) = (1 - \varepsilon_0)^H$. Equipping each atomic step with an independent spectral or invariant gate of false-pass rate $\varepsilon_{\text{gate}} \in [0, 1)$ and bounded retry budget $m \geq 1$ elevates reliability to:

$$R_{\text{checked}}(H) \geq [\, 1 - \varepsilon_0{}^m - \varepsilon_0\, \varepsilon_{\text{gate}}\, (1 - \varepsilon_0{}^{m-1}) / (1 - \varepsilon_0) \,]^H.$$

Derivation. Summing unscreened escape events $\varepsilon_0{}^{j+1}\varepsilon_{\text{gate}}$ across $j = 0, \ldots, m-2$ plus budget exhaustion $\varepsilon_0{}^m$ at attempt m bounds per-step error by $\tilde{\varepsilon} \leq \varepsilon_0{}^m + \varepsilon_0\varepsilon_{\text{gate}}(1 - \varepsilon_0{}^{m-1})/(1 - \varepsilon_0)$; independence across H gated cells yields $(1 - \tilde{\varepsilon})^H$. ■

Proposition 4 (Topic-Nullspace Attribution and Shift Invariance).

Let observations decompose as $x = Q\,[s; t] \in \mathbb{R}^{d_s+d_t}$ with orthogonal $Q \in O(d_s + d_t)$, structural invariants s determined by generator profile y, and topical semantics t with $\|t\|_2 \gg \|s\|_2$. For $V_{\text{topic}} = Q\,[0; I_{d_t}]$, the projected operator $W_{\text{attr}} = W_{\text{ridge}}(I - V_{\text{topic}} V_{\text{topic}}{}^\top)$ satisfies exact topical invariance $W_{\text{attr}}\, x(s, t) = W_{\text{attr}}\, x(s, t')$ ($\|W_{\text{attr}}\, V_{\text{topic}}\|_F = 0$) and zero population public-to-private topic-shift risk gap $R_{\text{priv}}(W_{\text{attr}}) - R_{\text{pub}}(W_{\text{attr}}) = 0$.

Derivation. Since Q is orthogonal, $(I - V_{\text{topic}} V_{\text{topic}}{}^\top)Q[s; t] = Q[s; 0]$, eliminating all dependence on t while preserving structural signal s identically across arbitrary topic shifts. ■

Proposition 5 (Controlled Specification Entropy and Effective-Rank Compression).

Collapsing synonymous lexical branching of multiplicity $M_{\text{eff}} > 1$ into canonical ASD-STE100 primitives [10] lowers conditional lexical entropy by $\Delta H \geq \log M_{\text{eff}}$ and removes orthogonal nuisance covariance $\Sigma_{\text{nuis}} \succeq 0$ from $\Sigma_{\text{nat}} = \Sigma_{\text{STE}} + \Sigma_{\text{nuis}}$, contracting ridge effective dimension $d_{\text{eff}}(\lambda) = \text{tr}(\Sigma(\Sigma + \lambda I)^{-1})$ by operator monotonicity and tightening zero-shot hypernetwork generalization error $(\sigma^2 / N_{\text{meta}})\, d_{\text{eff}}(\Sigma_{\text{STE}}, \lambda)$. ■

3. Next-Generation Weight-Space Model Families (7–10)

Building upon Families 1–6 (whitened nullspace 24D embeddings, hybrid RRF search weights, polytope Wanda routers [5], 256 B `LoRA-XS` cartridges, orthogonal memory-as-`LoRA`, and steered open profiles), we formulate four next-generation weight families:

- **Family 7 — Topic-Nullspace Generator Attribution (`TopicNullspaceAttributionCompiler`):** Across $C = 8$ structural generator profiles spanning four seen training topics and two unseen holdout topics, we extract a 64D structural invariant block (topic-masked character 3-gram sketches, sentence-length Fano factor $\text{Var}(L)/\mathbb{E}[L]$, 4×4 function-word Markov transition matrices, punctuation cadence, and token-length curvature) entangled with a 32D topical block via $Q \in$

$O(96)$ [3]. Within-profile cross-topic differences yield V_{topic} and $W_{\text{attr}} = W_{\text{ridge}}(I - V_{\text{topic}}V_{\text{topic}}^{\top})$, while conformal class-conditional nullspace residual energy screens adversarial style spoofing at 0.9866 AUROC.

- **Family 8 — Lifted Koopman Tool-Graph Actors (`KoopmanToolTrajectoryActor`):** Raw 8D states on a six-tool DAG with deceptive myopic traps lift via polynomial-bilinear observables [2] $\psi(s) \in \mathbb{R}^{32}$ into action-specific `float32` Koopman matrices $K_a \in \mathbb{R}^{32\times32}$ and linear feasibility gates $g_a \in \mathbb{R}^{32}$, planning 100.0% valid trajectories in 0.112 ms with zero LLM tokens.
- **Family 9 — Zero-Shot Spec-to-Cartridge Hyper-Compiler (`SpecToCartridgeHyperCompiler`):** Maps an ASD-STE100 specification vector $e_{\text{spec}} \in \mathbb{R}^{24}$ at Step 0 (zero task traces) via ridge hypernetwork $\Theta \in \mathbb{R}^{64\times24}$ into a 256 B (8×8 `float32`) `LoRA-XS` core R_0 in 0.0038 ms (35.59% → 85.34% zero-shot, reaching 98.67% by `CEGAR` Wave 3 [4]).
- **Family 10 — Spectral Screening & Legendre Bitemporal Forecasting:** `SpectralResidualGate` fits an orthonormal rank-12 trace basis $U_k \in \mathbb{R}^{48\times12}$ (1.0000 poison AUROC in 0.0012 ms/trace), while `LegendreBitemporalForecaster` projects queue-load and token-cost histories onto shifted Legendre polynomials $P_{0..6}$ under strict `valid_at ≤ t_cut` and `known_at ≤ t_cut` filtering (0 temporal leaks, −94.72% RMSE).

Table 1. Control 0 Monolithic Null-Effect, Granularity Phase Transition, and Prior Families 1–6 (5 Seeds {11, 23, 37, 53, 71}).

Study / Weight Family	Matched Baseline	Naive / Unprojected Edit	Atomic Compiled Cell (Invariant)
Control 0 (0.016% sparse edit)	51.84% (`thinking=off`)	51.60% (−0.24 pp, $t = -1.395$)	65.48% ($N = 1$ trace operator)
Granularity ($N = 1 \rightarrow 32$)	51.99% base	−1.26 pp ($N = 1$ monolith)	96.65% (+44.66 pp at $N = 32$; 98.96% sym)
ASD-STE100 Compression	62.36% (24D free text)	102.37 effective rank	100.00% (92.26 rank, −26.36% entropy)
96-Step Horizon Scaling	1.89% unchecked	Open-loop compounding	98.02% gated atomic pipeline
F1: Whitened Nullspace 24D	1.77% Recall@1	1.77% (0.033 drift)	96.67% ± 1.59% (90.6% RAM cut, 0.0 drift)
F2: Hybrid Asymmetric RRF	0.2528 nDCG@10	0.2445 nDCG@10	0.9429 ± 0.0246 (0.0046 ms)
F3: Polytope Wanda Router	17.01% accuracy	93.26% (3.68% rule viol.)	99.58% ± 0.38% (75% pruned, 0.0% viol.)
F4: Subspace `LoRA-XS`	8.75% accuracy	88.23% naive merge	99.01% ± 0.72% (256 B, 0.036 ms)
F5: Orthogonal Memory `LoRA`	35.55% (8,192 tok)	47.42% (87.5% interference)	100.00% ± 0.00% (0 tok, 0.0% interference)
F6: Steered Open Profile	27.92% accuracy	7.92% magnitude prune	100.00% ± 0.00% (50% FFN, 80.7% tok cut)

Table 2. Five-Seed Empirical Results for Next-Generation Weight-Space Families 7–10 (Seeds {11, 23, 37, 53, 71}, CPU).

Next-Gen Family	Matched Baseline	Unprojected / Myopic Control	Compiled Weight-Space Operator
F7: Attribution (8 profiles)	20.31% surface heuristic	95.00% seen → 22.34% ± 1.00% unseen	100.00% seen → 99.30% ± 0.38% unseen
Shift & Conformal Spoof Gate	N/A	72.66 pp topic-shift drop	0.70 pp shift, 0.0 leak, 0.9866 AUROC
F8: Koopman 6-Tool Actor	17.50% open-loop	16.25% ± 0.00% greedy 1-step	100.00% ± 0.00% (0 invalid, 0.112 ms)
F9: Spec-to-256B Hypernet	35.59% base (35.10% rand)	Zero traces at Step 0	85.34% Step 0 → 98.67% Wave 3 (0.004 ms)
F10A: Spectral Residual Gate	Unscreened stream	In-manifold projection only	1.0000 ± 0.0000 AUROC (0.0012 ms/trace)
F10B: Legendre Bitemporal	14.826 persistence RMSE	10.272 RMSE (1,000 leaks)	0.784 ± 0.137 RMSE (0 leaks, −94.72%)

4. Adversarial Red-Team, Non-Symbolic Operators, and Real Pretrained `.safetensors` Surgery

Adversarial Stress & `CEGAR-WSP`: Across eight targeted red-team attacks—contradictory polytope rules (6.55% → 100.0% via Farkas dual infeasibility certificates), rank-58 nullspace saturation in 64D (11.04% → 100.0% via Tikhonov slack + 64D → 96D auto-expansion at 0.0 Tier-1 drift), 25% poisoned traces (18.54% → 100.0% via disjoint lineage + Huber IRLS), collinear 256 B `LoRA-XS` merges (44.58% → 100.0% via Grassmannian QR), 2% rare-tail Wanda pruning (8.33% → 100.0% via upper-quantile CVaR), polarity twins in 24D (46.88% → 99.10% via bilinear projections), Wave-3 OOD inputs on 4,545-param routers (100% silent corruption → 0.0% via Mahalanobis energy gate), and Byzantine swarm merges (1.17% → 100.0% via Weiszfeld geometric median)—mean accuracy recovers from 17.14% to 99.89% (+82.75 pp). Over six refinement waves, `CEGAR-WSP` prevents −17.97 pp model-generated collapse (73.90% → 55.93%), reaching 99.73% ± 0.25% with 6.67× fewer oracle scenarios, while `AstTemplateWeightCompiler` synthesizes six reusable Python AST templates across `{Transfer, Transform, Store}` with 100.0% unit-test pass (0 LLM tokens, −92.89% wire bytes).

Non-Symbolic Spectral, Familiarity & Streaming Operators: On curved and episodic manifolds where symbolic rules do not apply, eight closed-form 0-epoch operators compile `float32` weights directly (Table 3): Bochner RFF [6] (99.22%), Nyström landmarks (98.06% at M = 48, beating 5×-larger D = 240 RFF by +10.95 pp), 1-shot empirical NTK updates (24.50% → 95.75% at 0.0 anchor drift), Modern Continuous Hopfield familiarity routing [7] (0.0045 ms 1-shot column writes, 100% novelty AUROC), Kanerva VSA holographic binding [8] (100% 3-slot unbinding under 30% corruption), 1-shot Fisher imprinting (40.59% → 100.0%), Recursive Woodbury RLS (1.5×10^{-13} batch-ridge exactness, 0.0 pp forgetting, 0 replay rows), and data-free Marchenko–Pastur RMT [9] + CKA deduplication (100% automated layer diagnosis).

Real Pretrained `.safetensors` Surgery: Across 11 real pretrained PyTorch `.safetensors` matrices (7.56M-parameter subword embeddings [29,528 × 256], 12-layer Vision Transformer [384 × 384] / [1536 × 384], and 768D sequence Transformer MLP blocks), Marchenko–Pastur bulk-edge diagnostics identify 114–268 signal spikes per matrix ($\alpha \in [2.20, 5.10]$). Nullspace ridge surgery on the 7.56M-parameter subword table compresses 256D vectors into a 6D cartridge (−97.66% RAM), lifting polarity accuracy from 50.0% to 100.0% at 2.30×10^{-16} anchor drift, while orthogonal Procrustes stitching restores 3-model cross-space Recall@1 from 0.0–6.25% to 100.0% with 0 vectors re-indexed.

Table 3. Non-Symbolic Weight-Space Operators and Real Pretrained `.safetensors` Benchmarks.

Operator / Pretrained Experiment	Baseline / Naive Control	Compiled Weight Result	Key Invariant / Cost
Adversarial 8-Paradigm Mean	17.14% under attack	99.89% (+82.75 pp)	0.0 drift, 100% poison quarantine
`CEGAR-WSP` (6 Waves)	55.93% self-train collapse	99.73% ± 0.25%	6.67× fewer oracle scenarios
Bochner RFF + Nyström (M = 48)	58.07% linear / 79.00% RFF	99.22% RFF / 98.06% Nyström	0 epochs, 5× smaller rank
Modern Hopfield Familiarity	0.0% 1-shot static router	100.0% recall & AUROC	0.0045 ms 1-shot column write
1-Shot Fisher + Woodbury RLS	40.59% cos / 38.44 pp forget	100.0% 1-shot / 99.11% 32-shot	1.5×10^{-13} diff, 0 replay rows
Real 7.56M Subword Surgery	50.0% slice (0.308 drift)	100.0% in 6D (−97.66% RAM)	2.30×10^{-16} anchor drift
3-Model Procrustes Stitching	0.00–6.25% cross Recall@1	100.0% across all 3 pairs	0 stored vectors re-indexed

5. Seven-Target Zero-Dependency Compilation & (ε, δ)-DP Nullspace Woodbury Merging

5.1 Seven-Target IEEE-754 Cartridge Portability (Including In-Memory `.wasm` and WebGPU `WGSL`)

Because compiled weight cells reduce runtime inference to compact `float32` matrix-vector products and piecewise-linear activations, a single cartridge executes identically across server, browser/edge WebAssembly [11], WebGPU compute shaders, embedded SQL, and bare-metal C99 substrates without framework lock-in. For supply-chain auditability, `WasmWgslKernelCompiler` synthesizes valid IEEE-754 WebAssembly v1 binary modules (`\0asm` v1 sections for types, linear memory, active data segments, and `f32.load` / `f32.mul` / `f32.add` / `f32.max` loops) strictly *in memory* as `base64`

streams alongside human-readable `.wat` and `@compute @workgroup_size` WebGPU `.wgsl` source, writing zero binary blobs to source repositories.

Table 4. Seven-Target Zero-Dependency Cartridge Equivalence (32 Test Vectors, 16 → 12 → 4 `float32` ReLU+Linear Cartridge).

Executable Target	Execution Substrate	Batch (32 vec)	Top-1 Match	External Dependencies & Artifact Form
1. `.safetensors` Binary	PyTorch / NumPy `float32`	0.281 ms	100.0%	IEEE-754 tensor loader (1,280 B)
2. ES Module JavaScript	Node V8 `Float32Array`	0.938 ms	100.0%	0 npm packages (`Math.fround`)
3. Pure Python Kernel	Python stdlib lists	0.396 ms	100.0%	0 external imports (`struct.pack`)
4. Relational SQL Query	`sqlite3 :memory: JOIN/MAX`	3.684 ms	100.0%	Declarative SQL dot-product
5. Static C99 Header	`gcc -O3` (`cartridge.h`)	0.0023 ms	100.0%	0.106 µs (106 ns) single-call
6. IEEE-754 `.wasm` v1	V8 / Browser `WebAssembly`	0.0184 ms	100.0%	In-memory bytecode + `.wat` (0 disk blobs)
7. WebGPU `WGSL` Shader	`@compute @workgroup_size(64)`	GPU parallel	100.0%	Storage-buffer `f32` compute kernel

Cross-Runtime Max Absolute Logit Discrepancy: $1.19209 \times 10^{-7} < 10^{-5}$ equivalence ceiling across all 7 targets.

5.2 (ε, δ)-DP Nullspace Woodbury Federated Swarm Merging

When M isolated worker enclaves assimilate local domain observations without sharing raw records, each enclave $m \in \{1, \ldots, M\}$ computes a local Recursive Woodbury weight delta $\Delta_m \in \mathbb{R}^{c \times d}$ strictly on nullspace-projected covariates $\tilde{x} = P_\perp x$, where $P_\perp = I_d - V_a V_a^\top$ annihilates shared anchor basis V_a. Under (ε, δ)-differential privacy [12], each enclave update is clipped in Frobenius norm to sensitivity bound $C > 0$, perturbed by calibrated Gaussian noise, and post-projected onto Null($V_a^\top$):

$$\overline{\Delta}_m = \Delta_m \cdot \min(1, C / \max(\varepsilon_{\text{floor}}, \|\Delta_m\|_F)), \qquad \Delta W_{\text{DP}} = (1 / M)\,(\textstyle\sum_{m=1..M} \overline{\Delta}_m + Z)\, P_\perp,$$
$$Z_{ij} \sim N(0, \sigma^2), \qquad \sigma = C\,\sqrt{(2 \ln(1.25 / \delta))} / \varepsilon, \qquad \|\Delta W_{\text{DP}}\, V_a\|_F = 0.$$

Because $P_\perp$ depends exclusively on public anchor invariants V_a, right-multiplication by $P_\perp$ is a deterministic post-processing map preserving (ε, δ)-differential privacy while guaranteeing *identically zero* anchor drift $\|\Delta W_{\text{DP}}\, V_a\|_F = 0$ even under isotropic Gaussian noise Z. Across four autonomous enterprise canary topologies (`gateway_finops_offload` , `isolated_vpc_enclave` , `bitemporal_audit_ledger` , and `ci_patch_accelerator`), this protocol sustains 0 exfiltrated raw customer records, 0.0 anchor and entity subspace drift, 88.00% promoted token offload, and 0.008–0.014 ms p_{99} local latency.

6. Reproducibility and Configuration Grid Governance

Every behavioral, capacity, topological, threshold, and regularization scalar outside {0, 1, −1, 0.0, 1.0} is externalized in declarative JSON configuration grids and enforced via `ZeroHardcodedValuesAstGate` and `DomainAgnosticNamingGate` . All workload regimes traverse the seven-stage `GridLedger` with disjoint maker and checker lineages and automatic boundary expansion. The complete hardened runtime ships as `omni-fabric-runtime v0.6.0` (17 tools, 10 hooks, generation-5 `NodeGrid` , DNS-pinned SSRF-guarded sync, zero committed binary payloads, and zero external Python dependencies).

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